TrailBalance

One hiker, one ration, simplex

Linear Programming · First progress presentation

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One moderate day hike

10 km
500 m ascent
4 hours assumed

One 70 kg adult
Mild conditions, marked route
No overnight stay or planned refill

Fixed kit 2.3 kg

Water 2.5 L ≈ 2.5 kg

Decision: the lightest ration that meets our stated food targets

Scenario and masses: assumptions. Essential categories: [R3, R4]. Water guideline: [R4].

Carrying allowance and personal inputs

Input	Evidence	This milestone
Body mass	General day-pack guideline	10% × 70 kg = 7 kg
Age	Older-adult load studies	No age multiplier
Sex / gender	Population-specific sex findings	No blanket coefficient
Height	Torso and hip fit matter	No height-to-load formula

The reviewed sources do not validate one personal capacity equation from these four inputs.

[R5] REI pack-weight guidance. [R6] REI pack fitting. [R9] Walsh et al., 2018. [R10] Vickery-Howe et al., 2024.

The ration data

Per serving	Tortilla t	Tuna u
Edible mass	50 g	74 g
Food energy	140 kcal	70 kcal
Protein	4 g	17 g
Available servings	6	2

Assumed ration targets: at least 630 kcal and 33 g protein. Servings are continuous.

Label data: [R7] Mission Restaurant Style tortillas, [R8] StarKist 74 g tuna pouch. Inventory and targets: assumptions.

The minimum-mass LP

minimize $M = 50t + 74u$

$140t + 70u \geq 630$ kcal

$4t + 17u \geq 33$ g protein

$0 \leq t \leq 6, \quad 0 \leq u \leq 2$

$4800 + M \leq 7000$ g

All available food weighs 448 g. The pack limit is redundant here.

LP formulation: project derivation using [R1, R2]. Targets and kit masses are assumptions.

A feasible maximization form

$y_T = 6 - t$ $y_U = 2 - u$

maximize $\zeta = 50y_T + 74y_U$

$140y_T + 70y_U \le 350$

$4y_T + 17y_U \le 25$

$0 \le y_T \le 6, \quad 0 \le y_U \le 2$

$y = (0, 0)$ means packing the full inventory. The slack basis is feasible.

Equivalent reformulation: project derivation. Course starting form and slack basis: [R1], physical pp. 3-8.

The professor's simplex convention

xB = B^-1 b zN = N^T B^-T cB − cN

1. Enter a nonbasic variable with z < 0.

2. Compute direction d = B^-1 a_j.

3. Take min(xB_i / d_i) over d_i > 0.

4. Exchange the leaving and entering columns.

Stop when all zN ≥ 0. This uses the handout's sign convention.

[R1] The Simplex Method in Matrix Notation, physical pp. 6-11. Supporting reading: [R2], Chapter 5.

Pivot 1: tuna removal enters

Initial zN = (−50, −74). Enter yU.

Basic	Value	Direction	Ratio
sE	350	70	5
sP	25	17	25/17 ← minimum
sT	6	0	Excluded
sU	2	1	2

Step = 25/17. Protein slack sP leaves the basis.

Exact calculations from the project LP using [R1], steps 1-5. Slack names: energy E, protein P, inventory T and U.

Pivot 2: tortilla removal enters

z_yT = −554/17. Enter yT.

Basic	Value	Direction	Ratio
sE	4200/17	2100/17	2 ← minimum
yU	25/17	4/17	25/4
sT	6	1	6
sU	9/17	−4/17	Excluded

sE leaves. Final y = (2, 1), with zN = (277/1050, 49/15) ≥ 0.

Exact project calculations using [R1]. Final reduced costs follow the lecture's sign convention.

The optimal ration

4 tortilla servings
1 tuna serving

274 g

630 kcal · 33 g protein
Pack total 5.074 kg
Planning margin 1.926 kg



Classroom LP result. Both ration targets bind. Pack capacity remains slack. Source data: [R7, R8].

Reference map

[R1] Course handout, 26 October 2026: matrix primal simplex

[R2] Matoušek & Gärtner, 2007: Understanding and Using LP

[R3–R4] National Park Service & American Hiking Society: essentials

[R5–R6] REI: general pack-weight and backpack-fit guidance

[R7–R8] Mission & StarKist: the exact food-label coefficients

[R9–R10] Walsh et al., 2018 & Vickery-Howe et al., 2024: load studies

Full citations and URLs appear in the appendix, speaker notes and references/REFERENCES.md. Accessed 12 November 2026.

Future work

Next mathematical milestones

Measured masses and packed volume
More foods and infeasible scenarios
Whole equipment as integer decisions
Shared gear and fair group allocation

Website direction

Trip details and available inventory
Packing list with source assumptions
Capacity margins and binding constraints
Clear explanations of infeasibility

Current milestone: a small, explainable LP with a complete simplex solution

Proposed next milestones. Website, whole-item selection and group allocation remain future work.

Appendix: full matrix form

x = (yT, yU, sE, sP, sT, sU)

A	yT	yU	sE	sP	sT	sU	b
Energy	140	70	1	0	0	0	350
Protein	4	17	0	1	0	0	25
T inventory	1	0	0	0	1	0	6
U inventory	0	1	0	0	0	1	2

c = (50, 74, 0, 0, 0, 0). Initial B = I4.

Appendix: dual pivot updates

ΔzN = −(B^-1 N)^T e_i s = z_j / Δz_j

Pivot	Pre-exchange N	ΔzN	Dual step s
1	(yT, yU)	(−4, −17)	74/17
2	(yT, sP)	(−2100/17, 70/17)	277/1050

Final nonbasic costs: z_sE = 277/1050, z_sP = 49/15

Appendix: an optimality certificate

$\lambda_E = 277/1050$ $\lambda_P = 49/15$ both ≥ 0

$140\lambda_E + 4\lambda_P = 50$

$70\lambda_E + 17\lambda_P = 74$

Every feasible ration has $M \geq 630\lambda_E + 33\lambda_P = 274$ g.

Our feasible ration attains 274 g, proving global optimality.

Appendix: fixed kit assumptions

Fixed group	g	Fixed group	g
Empty pack	650	Shelter + whistle	120
Navigation + backup power	300	Sun protection	80
Warm layer + rain shell	450	Repair / fire / waste	80
Headlamp + batteries	100	Water containers	150
First aid + medications	150	Emergency food	200
Food packaging	20	TOTAL	2300

All masses are assumptions. Water contents add 2500 g separately.

References: LP and hiking guidance

[R1] The Simplex Method in Matrix Notation. Course handout, 26 Oct 2026. Physical pp. 3–11, 12–18. Author unidentified.

[R2] Jiří Matoušek & Bernd Gärtner. Understanding and Using Linear Programming. Springer, 2007. doi:10.1007/978-3-540-30717-4

[R3] U.S. National Park Service. Ten Essentials. nps.gov/articles/10essentials.htm

[R4] American Hiking Society. The Ten Essentials of Hiking. americanhiking.org/10essentials/

[R5] Joe Pasteris, REI. How Much Should Your Pack Weigh? rei.com/learn/expert-advice/backpacking-weight.html

References: fit, foods and research

[R6] REI. How to Size and Fit a Backpack.
rei.com/learn/expert-advice/backpacks-adjusting-fit.html

[R7] Mission. Soft Taco Restaurant Style Flour Tortillas, label.
missionfoods.com/products/soft-taco-restaurant-style-flour-tortillas

[R8] StarKist. Light Tuna in Water (Pouch), 74 g label.
starkist.com/product/light-tuna-in-water-pouch/

[R9] Walsh, Low & Arkesteijn. Older-adult load-carriage study.
J. Biomech. 73 (2018), 18–23. doi:10.1016/j.jbiomech.2018.03.018

[R10] Vickery-Howe et al. Male/female military load-carriage study.
Biomechanics 4 (2024), 382–410. doi:10.3390/biomechanics4030027